

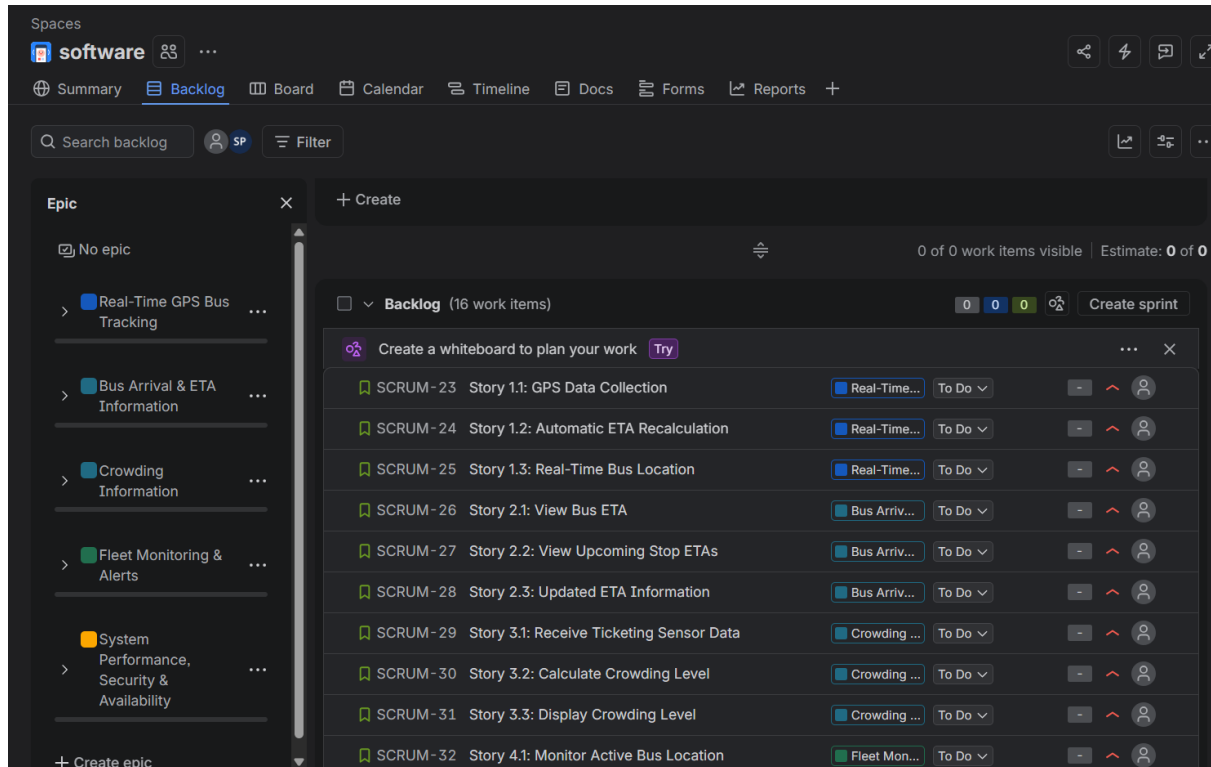
Lab 2: Agile Backlog Creation & Sprint Simulation in Jira

NAME: SPOORTHI.P

SRN: PES1UG24CS629

SECTION: K

EPICS AND STORIES:



Real-Time GPS Bus Tracking



▼ Description

Enables the system to receive GPS coordinates and track the real-time location of active buses

▼ Attachments

 [Add attachment](#) [Create project plan](#) [Try](#)

▼ Child work items

...  +

100% Done

Work	Pri...	As...	Status
 SCRUM-23 Story 1.1: GPS Data Collection	 H.	 U	Done ▾
 SCRUM-24 Story 1.2: Automatic ETA Recalculation	 H.	 U	Done ▾
 SCRUM-25 Story 1.3: Real-Time Bus Location	 H.	 U	Done ▾

Spaces / software / SCRUM-19

Bus Arrival & ETA Information

Description

Enables commuters to view accurate estimated arrival times of active buses at upcoming route stops.

Attachments

Add attachment

Create project plan Try

Child work items

100% Done

Work	Pri...	As...	Status
SCRUM-27 Story 2.2: View Upcoming Stop ETAs	H.	U	Done
SCRUM-28 Story 2.3: Updated ETA Information	M	U	Done
SCRUM-26 Story 2.1: View Bus ETA	H.	U	Done

STORY POINTS:

Spaces / software / SCRUM-18 / SCRUM-23

Story 1.1: GPS Data Collection

Key details

Priority

High

Description

As a **Fleet Controller**, I want the system to receive GPS coordinates from active buses every 5 seconds, so that I can track the current location of buses in real time.

Attachments

Subtasks

Add subtask

Linked work items

Add linked work item

Done

Done

Improve Story

Details

Assignee

Unassigned

Assign to me

Labels

None

Parent

SCRUM-18 Real-Time GPS Bus

Due date

None

Team

None

Start date

None

Story point estimate

8

Sprint

None +1

Reporter

SP Spoorthi Pathroti

Spaces / software / SCRUM-19 / SCRUM-27

Story 2.2: View Upcoming Stop ETAs

Key details

Priority

High

Description

As a **commuter**, I want to see estimated arrival times for upcoming route stops, so that I can understand when the bus will reach my required stop

Attachments

Add attachment

Create product requirements

Try

Subtasks

Add subtask

Linked work items

Done

Done

Improve Story

Details

Assignee

Unassigned

Assign to me

Labels

None

Parent

SCRUM-19 Bus Arrival & ETA In

Due date

None

Team

None

Start date

None

Story point estimate

5

Sprint

None

+1

Reporter

SP Spoorthi Pathroti

Spaces / software / SCRUM-20 / SCRUM-29

Story 3.1: Receive Ticketing Sensor Data

Key details

Priority

Medium

Description

As a **system**, I want to receive valid ticketing sensor data from active buses, so that passenger occupancy can be determined.

Attachments

Add attachment

Create product requirements

Try

Subtasks

Add subtask

Linked work items

Done

Done

Improve Story

Details

Assignee

Unassigned

Assign to me

Labels

None

Parent

SCRUM-20 Crowding Informati

Due date

None

Team

None

Start date

None

Story point estimate

5

Sprint

None

+1

Reporter

SP Spoorthi Pathroti

Story 4.1: Monitor Active Bus Location

Key details

Priority

High

Description

As a **Fleet Controller**, I want to view the live location of active buses, so that I can monitor the fleet effectively.

Attachments

Add attachment

Create product requirements

Try

Subtasks

Add subtask

Linked work items

Done

Done

Improve Story

Details

Assignee

Unassigned

Assign to me

Labels

None

Parent

SCRUM-21 Fleet Monitoring & A

Due date

None

Team

None

Start date

None

Story point estimate

5

Sprint

None

+1

Reporter

SP Spoorthi Pathroti

Spaces / software / SCRUM-22 / SCRUM-38

Story 5.3: Secure Tracking Data

+ ... ⇅

Key details

Priority

High

Description

As a **Fleet Controller**, I want bus tracking and telemetry data to be handled securely, so that sensitive operational information is protected.

Attachments

Add attachment

Create product requirements Try

Subtasks

Add subtask

Linked work items

Done Done ⚡ ⬆ Improve Story

Details

Assignee

Unassigned

Assign to me

Labels

None

Parent

SCRUM-22 System Performance

Due date

None

Team

None

Start date

None

Story point estimate

8

Sprint

None +1

Reporter

SP Spoorthi Pathroti

SPRINTS:

software

Summary Backlog Board Calendar Timeline Docs Forms Reports +

Search backlog

SP Filter

SCRUM Sprint 1

Add dates

(6 work items)

32 0 0

Start sprint

SCRUM-23	Story 1.1: GPS Data Collection	Real-Time GPS Bu...	To Do	8		
SCRUM-24	Story 1.2: Automatic ETA Recalculation	Real-Time GPS Bu...	To Do	8		
SCRUM-25	Story 1.3: Real-Time Bus Location	Real-Time GPS Bu...	To Do	5		
SCRUM-27	Story 2.2: View Upcoming Stop ETAs	Bus Arrival & ETA I...	To Do	5		
SCRUM-26	Story 2.1: View Bus ETA	Bus Arrival & ETA I...	To Do	3		
SCRUM-28	Story 2.3: Updated ETA Information	Bus Arrival & ETA I...	To Do	3		

+ Create

SCRUM Sprint 2

Add dates

(10 work items)

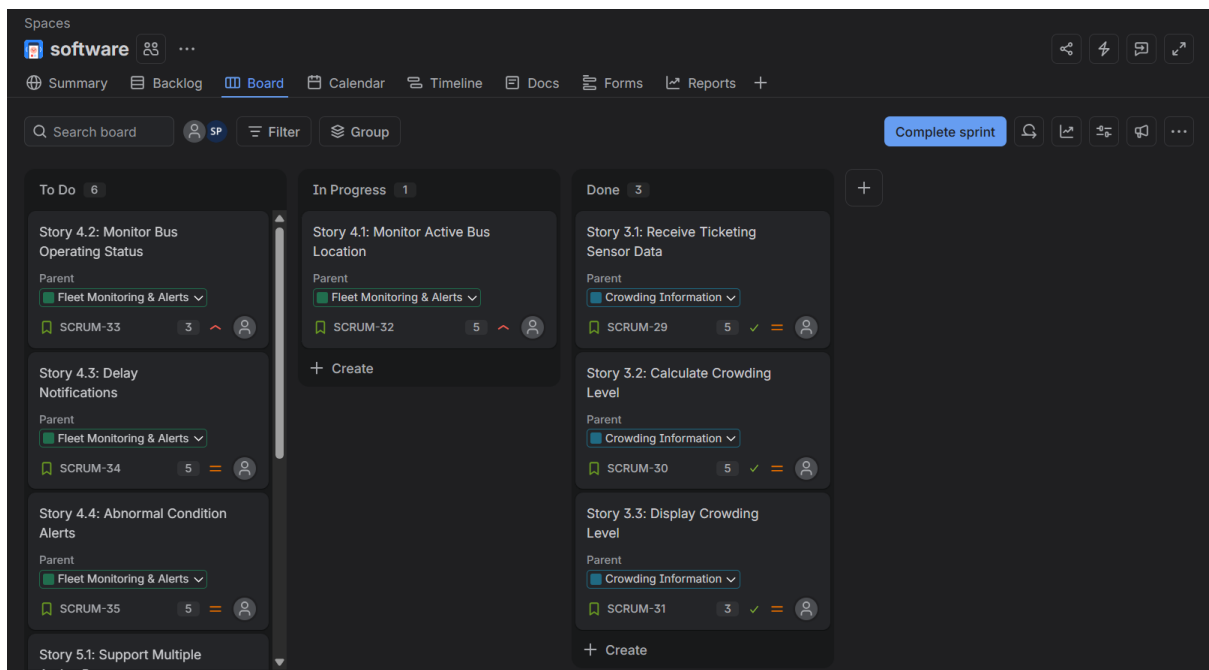
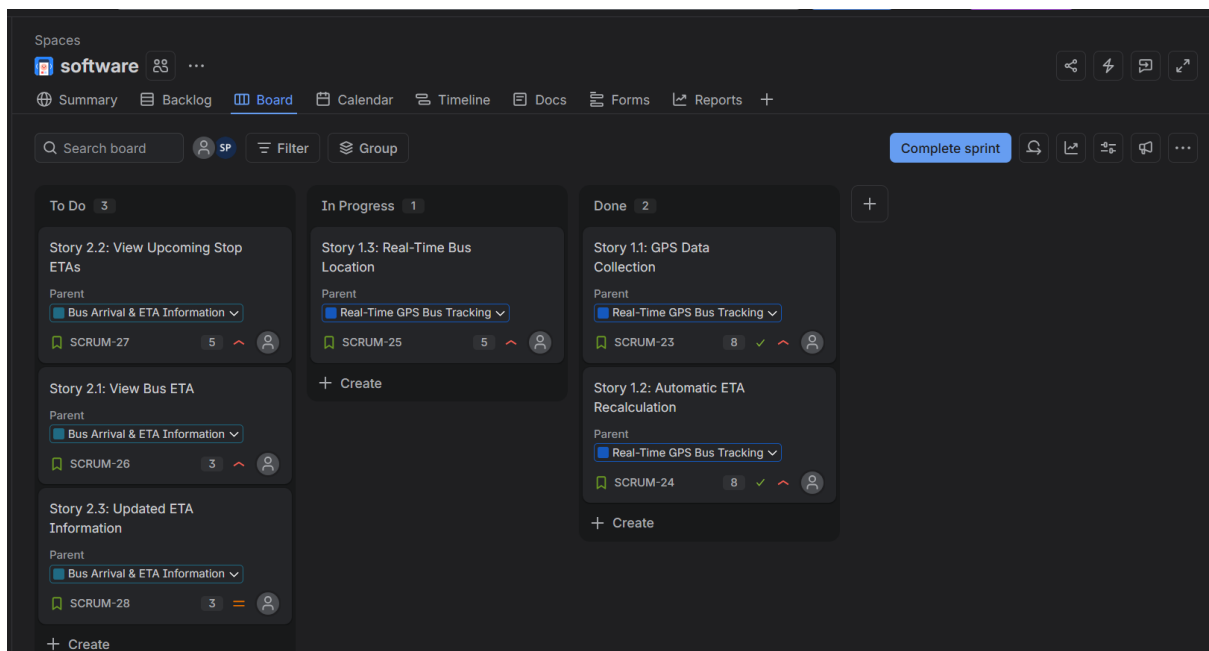
60 0 0

Start sprint

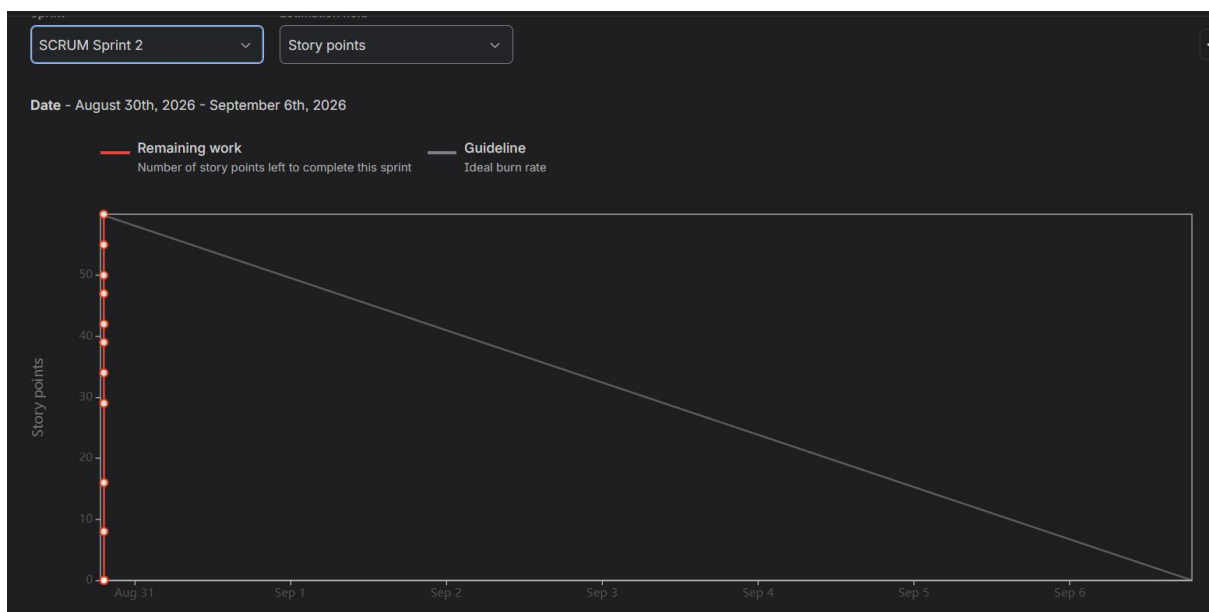
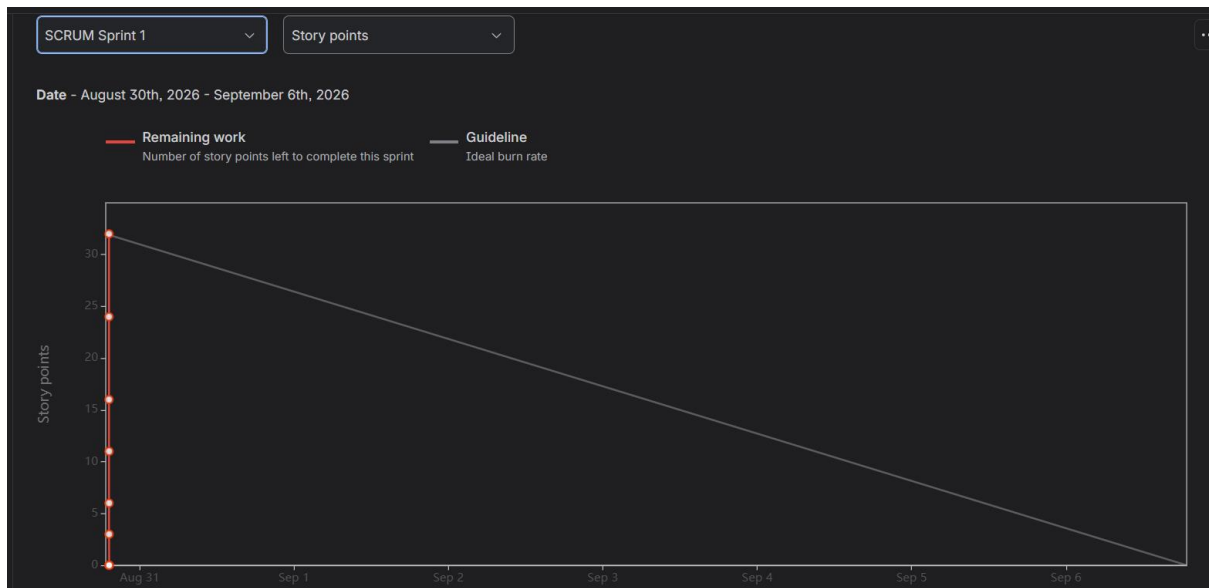
SCRUM-29	Story 3.1: Receive Ticketing Sensor Data	Crowding Informat...	To Do	5		
SCRUM-30	Story 3.2: Calculate Crowding Level	Crowding Informat...	To Do	5		
SCRUM-31	Story 3.3: Display Crowding Level	Crowding Informat...	To Do	3		
SCRUM-32	Story 4.1: Monitor Active Bus Location	Fleet Monitoring & ...	To Do	5		

6 of 6 work items visible | Estimate: 32 of 32

ACTIVE SPRINT BOARD:



BURNDOWN CHART:



Reflection Questions

1. Did your estimations reflect the actual effort?

Yes, the story point estimations generally reflected the relative effort and complexity of the user stories. More complex tasks, such as GPS Data Collection, were assigned higher story points, while simpler features, such as viewing the estimated bus arrival time, were assigned fewer points. Since this was a simulated sprint, the estimations were used mainly to understand and compare the effort required for different tasks.

2. Was your backlog well-prioritized?

Yes, the backlog was prioritized based on user impact and system importance. High-priority features such as real-time GPS tracking, ETA display, crowding estimation, and fleet monitoring were given priority because they are essential functions of the system. Delay and

abnormal-condition alerts were given medium priority because they depend on the availability of monitoring data.

3. How did your simulated sprint align with your plan?

The simulated sprint mostly aligned with the planned work. The selected user stories were moved from **To Do** to **In Progress** and finally to **Done**, following the Scrum workflow. This demonstrated how work progresses during a sprint and helped simulate the completion of planned tasks within the one-week sprint duration.

4. What insights did the burndown chart give about your team's capacity?

The burndown chart showed how the remaining story points changed during the sprint compared with the ideal guideline. It provided an understanding of the team's capacity to complete the planned work within the sprint period. In this simulated sprint, the chart also showed that completing multiple stories at the same time can create a sharp drop rather than a gradual decrease. This indicates that, in a real project, work should ideally be completed progressively throughout the sprint.